Automatic Waffler

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PHYS 124: Laboratory Projects

Introduction: Everybody loves waffles. We propose the first device of it's kind to automate the cooking process of waffles for your eating convenience. The project will build upon the skills learned in PHYS 124, to design an effective and delicious solution to fully automate waffle making. We will use various motors, sensors, Arduino Software, and plenty of waffle batter to design the ultimate *Automatic Waffler*!

I. Motivation

It is estimated that 164 million Americans eat waffles every year, with the average person consuming at least 10-20 waffles per year. Despite the large presence of manufactured waffle makers in the US, evidence has yet to be shown of manufactures creating devices to automate the process of waffle making. The average time a waffle takes to cook is between 4 to 7 minutes. This suggest that given an efficient automation of waffle making, one may save over a billion minutes of personal cooking time from Americans. Furthermore, we project that injuries due to waffle making would dip from some small (order of 10) to effectively zero. Thus, through the most proper motivation, we demonstrate for our PHYS 124 project the first-of-its-kind automatic waffle maker.

II. Design

The design of the waffle maker is centered around a surrounding frame of 80/20 aluminum extrusion which supports all components of the waffle maker. We chose 80/20 to construct the frame out of because it offered a lot of rigidity and structure as well as being very modular and easy to attach components to. We connected pieces of 80/20 using pre-made 80/20 nuts which M5 screws attached into. We also used several pre-made 80/20 joints to create the general frame of the structure. In figure 1 we show how the 80/20 was constructed. We followed this frame a guide for when we constructed the frame. We followed the dimensions and the frame worked well to hold the waffle components.

The sensing component we used in for this project was a load cell. We use the load cell to create a functional scale, which in turn was able to actively sense the weight of the batter which would be deposited on the waffle iron. Notice how in figure 2 the load cell has a hollow cavity in it. This cavity ensure that the stress the load cell feels are in a repeatable spot when it is under load. The electrical components of the load cell will be discussed later.

Also notice how in 2 there is a part of the scale dedicated to attaching the scale to the 80/20. This is what the front view showing the rectangle with two cutout holes is showing. Now this is a important feature because it is what centers the scale under neath the bowl suspending batter above it.

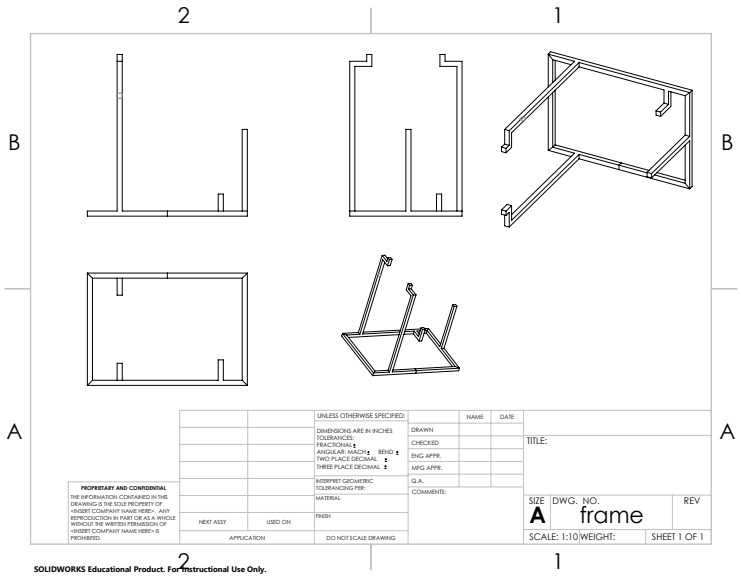


Figure 1: Model of how the frame was designed in CAD. We implemented this structure of the frame exactly, with all parts constructed out of 80/20 aluminium extrusion. Overall, this gave a very sturdy and dependable frame construtions.

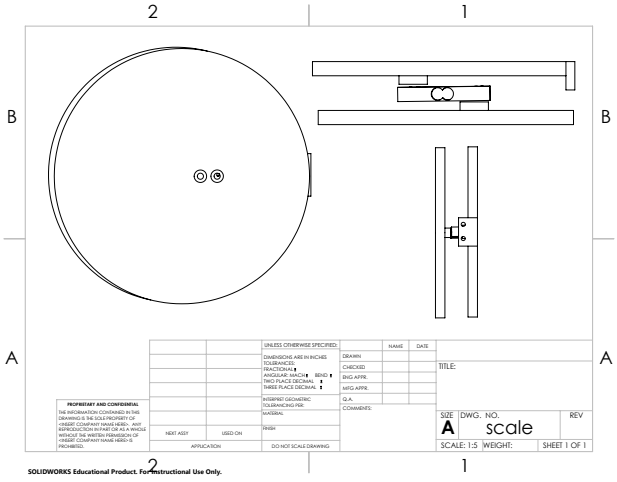


Figure 2: This is the design of the scale we used. In practice we actually used a square top plate and bottom plate to the scale, however the design was practically identical. In the scale we can see the shape of the load cell we used, and how it was put under stress to measure forces.



Figure 3: Here is a picture of the valve that we use. With no supplied voltage the valve sits in the normally closed position.

The valve we used for dispensing waffle batter was a 3/4 inch solenoid valve, which we directly mounted to the bottom of a plastic bowl which I had lying around my house. In 3, the valve has pipe threading of 3/4 inch. We then threaded in small piece of metal tubing to between the valve and a adapter which we where able to attach to the bottom of my bowl easily.

The metal attachment piece with the metal pipe holding the solenoid in figure 4, has screw holes which we use to thread M5 screws through and bolt them down on the other side of the bowl. This attachment piece then needed to be water tight, so we used 5 minuet epoxy to seal any holes. These materials where easily found at a hardware store.

The next part we designed was a capstan with we attached to the end of a stepper motor to pull a piece of para cord and hoist the lid of the waffle maker. We made this piece out of wood in the wood shop and we designed it in solid works. These parts are demonstrated in 5 and 6.

This pulley system directed rope upwards and over a custom 3d printed wheel that we made which attached to the 80/20. This is shown in figure 7

The last thing we did is we put all these above parts together and created the whole



Figure 4: This is the bottom of the bowl with the attachment we gave to it. The solenoid simply threads into the bottom of the bowl, and the threaded metal pipe attaches it to the bottom of the bowl.

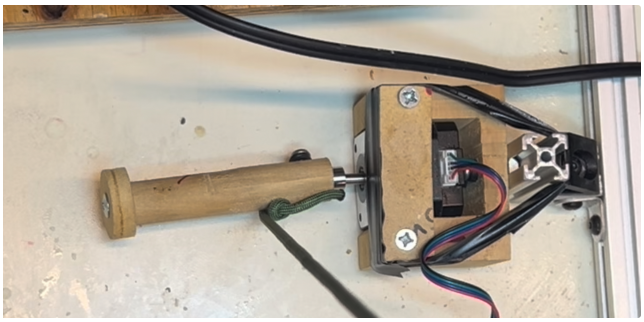


Figure 5: Woden Capstan that is attached to the stepper motor. We can also see that is winding some of the rope which winds to the waffle iron.

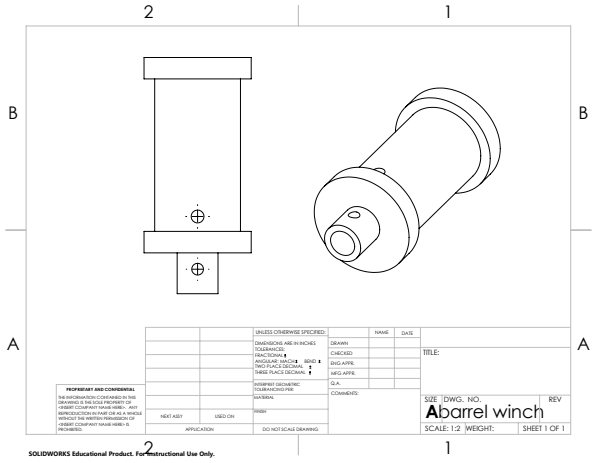


Figure 6: This is the a drawing of the CAD model we based the capstan off of. We can see that we slightly modified how we constructed the part, however it mostly is identical.

system. In figure 8 we do not show that the waffle iron we used sits on the scale or the valve at the bottom of the bowl, but the rest of the parts are in.

The waffle iron we used was an amazon basics waffle iron which we modified. The modification we made was drilling a small whole in the handle to allow the para cord to pass

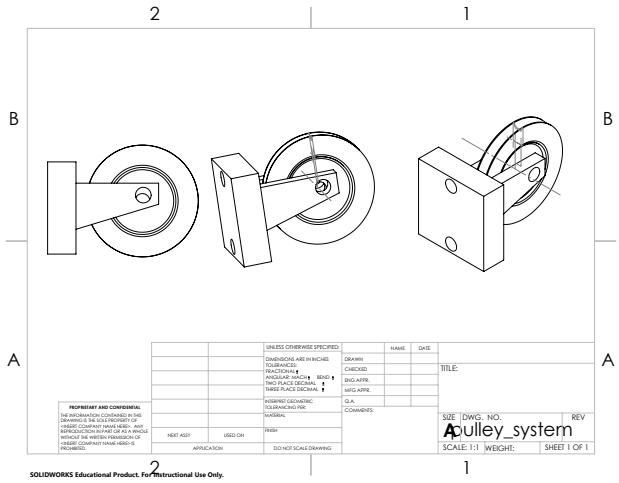


Figure 7: This is the custom wheel we made. It was held together with an M5 screw and a nut. This was attached to the 80/20 with some screws.

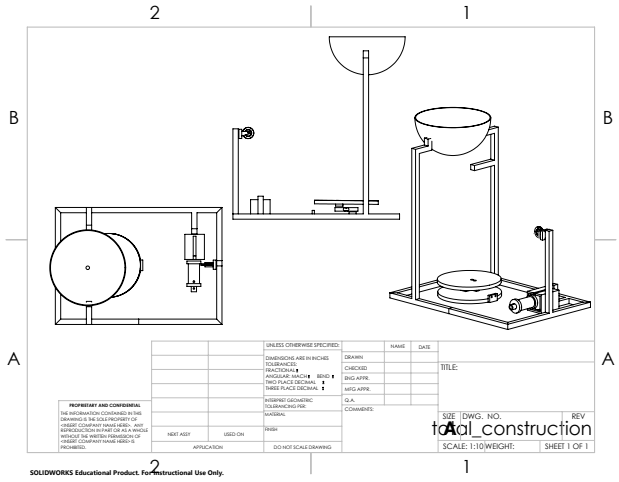


Figure 8: This is the assembly of all the parts we designed. Ultimately the waffle iron would rest on the scale.

through and allow the stepper motor to hoist the lid upwards. The waffle maker we used is in figure 9.

Ultimately the total construction used all the parts above. It can be found in figure 10



Figure 9: This is an image of the waffle maker we used



Figure 10: Everything put together

III. Functional Definition

The Automatic Waffler will delivers the following performance:

Once the batter reservoir is properly filled and the mechanism is powered, all a human needs to due is press the button, and apply PAM during the apply pam phase. Pushing the button will start the waffle making cycle. The machine will dispense a consistent and appropriate amount of waffle batter into a waffle iron. It will then shut and cook for a set amount of time. Once the waffle is done cooking it will open back up and be ready to be served.

Each of these steps require a different problem to be solved. We will use an Arduino digital input to check if a button has been pressed. We will use a force sensor to determine how much waffle batter has been placed onto the waffle iron. This will message to the dispenser to stop pushing out batter. We will generate the motion of the desired parts using stepper motors and controlling them using Arduino commands.

Ultimately the Automatic Waffler has 5 main stages to implement it. Stages 2 through 5 are done on repeat.

- Stage 1: Setup
  - Power on components.
  - Tare the scale
  - Calibrate the Scale
- Stage 2: Opening the Waffler
  - Press Button
  - Waffler lid opens
  - Ten second delay to apply PAM. This prints on serial monitor
- Stage 3: Batter Dispensing
  - Solenoid valve is turned one.
  - Scale measures weight change and shuts valve when 70 grams of batter has been deposited
- Stage 4: Cooking
  - Waffle iron shuts for 4 minuets
  - Opens back at the end of the timer
- Stage 5: Serving
  - Waffle Iron opens for 20 seconds
  - Grab the waffle in this step
  - Iron closes and is reset for another waffler

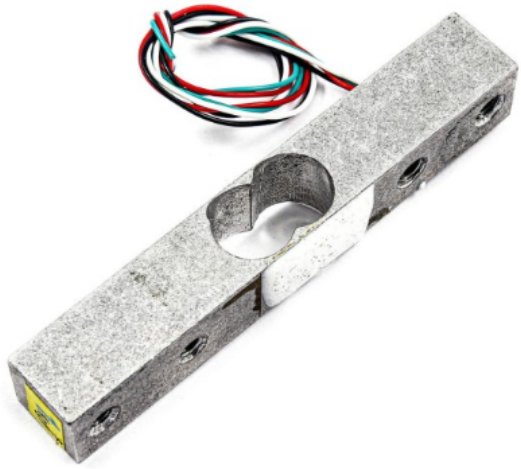


Figure 11: Image of the load cell we used. Notice the location of the cutouts

IV. Sensors

The main sensing we will be doing will be measuring the change of weight which the waffle iron feels as batter is deposited onto it. Our goal is to have this sensor be as reliable as possible so we get a very consistent amount of batter of the waffle iron before each cook. To do this, at each dispensing stage, we will write a code to tare out the weight of the waffle iron and only measure the change of the weight of the batter being deposited. Ultimately this sensing mechanism is the most crucial aspect of the project because it will control when and how the other parts to the mechanism will operate. These mechanism will be thoroughly laid out it the Mechanical Considerations section of this proposal.

Supporting this sensor will be no small task, as we will need is reliably in the same location, while also accurately measuring the batter being dispensed. We will build a housing and fix it to the frame of the machine to insure these qualities. This housing will need to be both rigid. The rigidity will come from where we place it

We used a 10kg load cell as our force measuring instrument. This load cell readily available online and has a useful connection to a HX711 amplifier. The combination of the load cell and the amplifier makes for simple integrating into the Arduino.

The load cells works via the use of a Wheatstone bridge. The Wheatstone bride is used by having 4 strain gauges form a bridge in a circuit. The load cell then outputs the voltage change and the HX711 amplifier converts this to useful information. Notice in figure 11 the load cell has specified cut outs in it. This is so that when a load is applied, the load cell deflects in a reliable and understandable location. The Wheatstone bridge is formed across the removes part of the load cell.

We used the HX711 amplifier using a package ¡HX711 Amp.h¿ which we found online. The Arduino and the amp communicated via serial.

The function of this sensor would be to measure the weight of the waffle iron as we deposit batter into it. Once enough of the batter has gotten into the waffle iron, the Arduino could determine the change in weight and accurately stop the flow of batter. In essence, we are

going to try and make a scale to measure how much batter has been deposited.

V. Mechanical Considerations

The Automatic Waffler is comprised of two main types of design features on it. First are the static components of the mechanism, and the second are the dynamic features.

Static Components

The static components consist of the frame of the machine, the batter reservoir, pressure plate stand, and the supports of the dynamic components.

Frame

The frame will need to be designed for a both supporting the heavy load of the batter reservoir and the resisting the forces which the dynamic components will create during motion. This frame will also need to be large enough to hold the batter reservoir high enough to avoid it colliding with the waffle irons lid opening. Further, we will hope to keep the manufacturing of the frame at a minimum to allow ourselves more time to create the other key components. Also having a cheap materiel to construct this frame would be a great convenience.

Keeping all these in mind we will use 80/20 aluminum extrusion which can be easily manufactured to our desired shape. We just used the horizontal band saw at the DIB. Further 80/20 has considerable amounts of pre-fabricated joints that are easily attachable, making the frame construction simple.

Using 80/20 also eliminated the fear of not having a frame that could support a heavy batter reservoir above the waffle maker. The 80/20 extrusion is very strong and offers ease of making robust parts. Using its pre-fabricated joints also makes attaching our custom components simple, as all we need is M5 screws and an allen key.

Batter Reservoir

The next static component that must be considered is the batter reservoir. This the feature will attached to the frame and held above the waffle iron. It will be held high enough such that fully opening the waffle iron will be possible without colliding with the reservoir. The reservoir will serve to hold the pre-mixed waffle batter and have the batter funnel towards the dispensing mechanism which will be described in the Dynamic Components section.

The construction of the batter reservoir came from modifying a plastic mixing bowl which I have at my house. The modifications include a design on how to attach the bowl to the 80/20 frame and a how attach the batter dispensing mechanism. For attaching the reservoir, we drilled through the bowl and bolted it into the 80/20 frame using M5 screws. For attaching the dispensing mechanism we made a small circular cut with the drill press. Then we will screwed through the bowl and attached a metal puck with 3/4 inch threading to it. We again used M5 screws to attach the metal puck, because the puck had designated screw holes. We then epoxied the bottom of the bowl to ensure water tightens. The reason

for the 3/4 inch threading in the puck is because the Valve we used for dispensing used a 3/4 inch threaded pipe. So we then used a small adapter to connect these two components.

Ultimately this bowl needed to be modulary attachable to the frame and the valve, while being water tight. We demonstrate the security of our bowl as a sufficient batter reservoir in our Video.

Pressure Plate / Force Sensor Stand

For our pressure plate stand, we cut out two rectangular pieces of wood singly connected to the 10kg LoadCell via centered screws. This allows the LoadCell to sit flush beneath the top plate of wood, with the bottom plate supporting from beneath. This allows us consistently and properly read a level force of the weight of batter as it dispenses onto the Waffler.

Using the bandsaw, we designed two "tetris" S and Z blocks that were then used to bracket the bottom wooden plate securely to our 80/20 frame. We attached one side of the bracket to the wood via wood screws and then connected the other side of the brackets to the 80/20 frame using long M5 screws and 2 nuts. It was of utmost importance that we correctly measured the bottom plate to be on the same level as the entire frame when it connected to its wooden brackets, to prevent any force slippage.

Dynamic Components

There will be two main dynamic components in this mechanism: the waffle opening winch and the batter dispenser.

Button

We used an in class push button to begin the waffle cooking process. This was simply connected over the Arduino pins with no extra components. These buttons where redly available.

Opening Winch

In order to ensure a secure opening and closing process for the waffle iron, we must have a secure winch. We opted for a winch opening mechanism because it trades off the high-speed, low force power of the stepper motor, for a slow speed, high torque driver, which is ideal for fine-tuning control of opening the Waffler. The entire opening winch is composed of a stepper motor with a wooden barrel attachment, the rope connecting the Automatic Waffler connected to the barrel, and a pulley. This also allows us to wind and unwind the rope in a repeatable manner. The wooden barrel redirects the rotary motion of the motor to a force which pulls the rope down when the waffle maker is opening, which then pulls the lid away of the waffle maker away from its base.

We secured a minimally sized pulley to the top of a long bar of 80/20. We eventually chose for a height on the order of the height of the batter reservoir, which gave us smoother transitions between opening and closing our waffle maker. A single-braided green rope was then secured to the capstan, with the other end burned and knotted secured to a hand-drilled hole at the frontal opening lip of the waffle maker.

As discussed above, the wooden capstan was instrumental in allowing the rope to continuously wrap around as it pulled the waffle iron open. We added a circular wooden cap to the end so the rope could continue piling along the barrel. The capstan can then be attached to the stepper motor itself with a single screw, allowing an efficient transfer of one torque to another without issues.

For the motor itself, as seen in Figure 5, we designed a secure housing which kept the stepper itself in place. Without this housing, the motor will try to balance the force it is applying onto the wooden capstan by turning in place. Thus we strapped the motor from it's sides with woods to prevent any turning. This wooden strap was then secured to the 80/20 frame leveled with the floor of the frame itself. 2 wooden screws were used to roof the top of the motor onto the strap. Another two screws not seen in figure 5 aligned perpendicular with the "normal"- pointing 80/20 connected via 2 more nuts secure the backboard to the frame. That little bugger is moving nowhere!

Dispensing Mechanism with Waffle Batter Considerations

A reliable interface between the batter reservoir and the dispensing systems is essential for controlling the batter flow in a repeatable way. A lot of consideration was made in thinking how to properly design the dispensing mechanism. We needed to consider how we could regulate the "flow" of a fluid that could allow for consistent fluid transfer.

As discussed in the Batter Reservoir section, a circular annulus is attached the bottom of the plastic bowl via two screws, which have a durable 5 minute epoxy covering the connections. The interface also has more epoxy inside the interior channel, which in totality forms a straight pipe with no holes. This prevents any leakage of the batter from outside the Batter Reservoir.

We then attached a 2 inch long, 3/4 inch diameter thick metal pipe to the annulus, where we used more epoxy to seal the internal screwing. The other end then attaches to the solenoid as depicted in figure 4. The distance then from the bottom of the solenoid valve, to the maximum height the waffle lid can reach was chosen to be as minimal as possible, to allow the waffle lid to clear the bottom of the valve. This then allows the fluid to drop and hit a target point (the center of the waffler maker) to prevent any unnecessary spilling of waffle batter. Because the batter *will* grab onto one of the sides of the valve, we angled the batter reservoir ever so slightly to offset the angle at which the batter will fall down.

The solenoid valve, now positioned inline with the batter reservoir, then can be electrically actuated by an the Arduino Uno, which then allows us precise on/off control of batter dispensing. The basic control logic relies on reading in the force from the load cell. When the waffle making process starts, the waffler iron will open, and stays open until the pushbutton is pressed. When pressed, the solenoid valve opens, dispensing batter. When the right "force threshold" is passed (aka enough batter has dispensed), the valve then gets closed.

What are we actually dispensing? These parts only form half of the dispensing mechanism. The fluid we are transporting is waffle batter, which has the property that it'll solidify after a few minutes of sitting in place.

In order to ensure the most laminar flow (with gravity pulling our batter down) we mixed the batter with enough water to have an almost liquid texture. We then "homogenized" the batter mixture, by straining the watery batter mixture through a fine-grated strainer.

This allowed us to remove any excess clumps which may possibly lead to jams inside the solenoid valve as the batter dispenses. We used the fact that the runny fluid (while having a reasonable weight) could dispense onto the waffle maker without splashing, and that it could evenly spread spherically from the target point, so it could occupy the entire waffle iron (timed with the force threshold). This behavior can be observed in our video.

Overall the Dispensing requires a sealed, rigid body, which must have a homogeneous mixture of waffle batter with a runny consistency, to ensure the smoothest flow of waffle batter onto the waffle iron.

VI. Electrical Considerations

The *Automatic Waffler* will require multiple electrical systems working in conjunction to actuate, sense and coordinate the major components of the whole device. To interface the mechanical components of the Waffler, we must **automate** batter dispensing, **actuate** opening and closing the waffle iron, and **detect** proper force measurements.

Thus, there are three main components of the Automatic Waffler, that will need a majorly connected electrical complex to operate efficiently and handle multiple tasks at once. Each system will have to be connected in parallel via one **Arduino Uno**. Via free digital and analog inputs, we will also have to use a motor shield to properly control the stepper motor if we choose to implement it. The following major mechanical systems need electrical considerations:

- Dispensing Mechanism
- Opening Winch
- Pressure Plate with Load Cell

Dispensing Mechanism

The *Dispensing Mechanism* has a Power Relay by Songle (Serial Code: SRD-05VDC-SL-C) which helps power the Solenoidal Valve. The valve requires a relay to properly gate the voltage as it switches the solenoid from being on/off. Connected to the relay through the COM is a constant 15V at 0.5A from the same power supply and the GND and IN terminals are connected to the Global Ground. VCC is connected to the same +5V from the Arduino and then NC is connected to the Arduino controlled output. Pins 2 and 3 switch whether the voltage reaches the solenoid. If it does it opens, otherwise it remains closed. No library is needed here.

Opening Winch

The *Opening Winch* is driven by a Stepper Motor bought off of MybotOnline (Serial Code: 17HS15-1704S). This motor is then connected to the L298N Motor Shield via its Out1-Out4 Motor Pins. An external power supply supplies a power at 18V and 1.5A connected to the

VCC 12V terminal on the Motor Shield. IN1-IN4 are connected to the Arduino via pins 8-11 controlled entirely via Stepper.h library. The +5V pin is connected to a +5V wire from the Arduino, and the GND terminal is connected to the global ground (external power x Arduino ground).

Pressure Plate with Load Cell

The *Pressure Plate with Load Cell* is connected to the HX711 Amplifier, for which we can read meaningful Serial Communication of the force output from the 10kg LoadCell, using the HX711.h package. The HX711 amplifies the micro voltages felt through the small resistors inside the Load Cell's Wheatstone Bridge, by which we can read out to our Serial Monitor.

The HX711 is connected to the Load Cells Wheatstone Bridge. It has 4 external pins GND, DT, SCK, and a VCC. The VCC is connected to +5V, and the GND to the global ground. The DT and SCK pins are connected to Arduino Pins 4 and 5 respectively, which is used in the HX711 constructor via the package in its arguments, so the package takes care of these pins.

VII. Software and Pseudocode

All programming will use the Arduino IDE in C, which will operate similarly to a **state-based controls system.** That is, steps will be progressed through iteratively to make waffles in a controlled, yet repeatable way which will depend on the previous task in conjunction with the various (tracked) waffle parameters. Additionally, as these tasks depend on the state of the other complexes, we concluded after speaking with Prof. Arnold that the most natural setup for implementing a state-based style approach only needed one Arduino Uno.

Using the Stepper Library, we fine tuned motor control for properly opening and closing the Waffler. To control the motor, we attached it to the motor shield and powered it with the power supply, then used the library. This is simply done and following the code should make this obvious.

The trickiest part of implementing the software is forming a cohesive waffle making process that can at least make three waffles without fail. Because their is a lot of interaction between both moving and static parts, which makes a lot of mismatching issues between the different Arduino's output values when changing system states.

Libraries required:

- Stepper.h (for Servo/Stepper Motor Controls)
- HX711 (to configure the load cell)

Additionally, we used this generalized pseudocode map, to track our waffle making process:

```
#include <HX711_ADC.h>
#include <Stepper.h>

HX711_ADC LoadCell(HX711_dout, HX711_sck);

int i= 0;
int steps = 700;
const int test_weight = 70; ;
unsigned long  wait = 240000;
int weight;

void setup{
// make all set up commands
}
void loop() {

new_state = digitalRead(button);


if (new_state != old_state && i !=0){

  weight = 0;
  Serial.println("open waffler");
  LoadCell.refreshDataSet();

  myStepper.step(-steps);  // One full rotation clockwise
  delay(1000);

  Serial.println("Apply PAM");
  delay(10000);
  Serial.println("Opening Valve");
  delay(3000);
  open_valve();
  if (LoadCell.update()){
    weight = LoadCell.getData();}
  Serial.println(weight);
  while (weight <test_weight){
    if (LoadCell.update()){
      weight = LoadCell.getData();}
      Serial.println(weight);
    }
```

```
  close_valve();
  delay(5000);
  myStepper.step(steps); // One full rotation counter-clockwise
  delay(wait);
  myStepper.step(-steps); //open back up
  Serial.println("Grab Your Waffle");
  delay(20000);
  myStepper.step(steps); // close and wait for another waffle


  // receive command from serial terminal
  if (Serial.available() > 0) {
    char inByte = Serial.read();
    if (inByte == 't') LoadCell.tareNoDelay(); //tare
    else if (inByte == 'r') calibrate(); //calibrate
    else if (inByte == 'c') changeSavedCalFactor(); //edit calibration value manually
  }

  // check if last tare operation is complete
  if (LoadCell.getTareStatus() == true) {
    Serial.println("Tare complete");
  }

}

old_state = digitalRead(button);
if (i ==0){i = i+1;}
}
```

VIII. Testing

Hardware

We needed to have a strong understanding on many of the Waffler's components to have a good build of the machine. To gain this understanding we tested pretty much all of the mechanical parts.

First, for the frame we tested how strong the 80/20 frame can be in supporting the batter reservoir. We needed to use a lot of brackets that held all of the major hardware components down onto the frame. We tested to make sure the metal brackets connected flushly against the frame aligned with the ground. For each electrical component, as seen in our video, we have a separate mini-BreadBoard to create a modular system. These also needed to be

covered with paper towels in order to prevent any possible batter from breaking our electrical components.

The Batter Reservoir alongside the dispensing system, needed a lot of testing in order to find the right consistency of batter. We had to ensure it was waterproof (which it wasn't initially), so we had to epoxy most of the spout. We then had to test the height of the entire dispensing mechanism to make sure the waffle lid cleared it minimally. Originally we had it too low so we had to add more 80/20 to increase the height of the Batter Reservoir. We then noticed that the angle at which the batter came out was off center from where we placed the waffle maker atop of the scale, so we needed to rescrew the Batter Reservoir at an angle to fix that offset.

The opening winch needed to also be tested a fair amount as we kept having issue with the motor stalling. After we fixed the stalling issue, we notice it wasn't driving the right amount of steps, so we needed to configure the correct amount of rotations of the motor to completely open the waffle iron with no issues. After we configured the right amount of steps at a convenient speed, we had problems with the rope exiting the wooded barrel, so we screwed on an additional circular cap. Finally, the motor continued to remain unstable until we built the wooden housing we secured it properly to the frame.

Software

In terms of software, by the design of our *Automatic Waffler*, we believe it would be relatively easy to manage and track where any issues may arise if it is a problem with our code. Each major mechanical part of our design is linearly connected to the Arduino Uno which can help us test the performance of our code individually for each mechanical part, and isolate which complex the problem is arising from. Perhaps it would be useful to implement checking procedures for each major electrical component needed, and can then be sent to the state machine to verify all systems to go (something like an automatic check).

While there are many moving parts (for both software and hardware components), the natural modular design of the *Automatic Waffler* should give us a nice advantage when testing out how the software matches Servo motor control, the valve openings and timings, and all of the simpler waffle parameters before entering COOKED state.

IX. Safety Measures

The Waffle Iron itself will be the main cause of concern for a safety hazard. The high temperatures can cook a waffle and burn skin or even worse, burn an Arduino! In order to be safe, whenever we operate the Waffle Iron, we will ensure that both lab partners have insulated gloves. In the frame, we aksi ensure that the waffle iron stays far from the electrical heavy components of the Project.

Although the Batter Reservoir may not necessarily be a safety hazard, it is still hard to control the flow of batter when testing out the solenoidal valve, and when depositing batter

onto the hot waffle iron. Thus we covered most of our component and wires with tissue paper, when going through test trials of waffle making.

The rest of the other major components of the project do not (yet) seem to be the cause for any concern. No high voltages, sharp objects, chemicals or other dangerous will be used.

X. Parts Required and Reusability and Shopping List

We used various adhesives, screws, and bolts as they come up zs they come up. We used ALOT of 80/20 parts, M5 screws with corresponding M5 sized nuts that are designed to fit into the interior of the 80/20. We also used a small amount of wood when making the Pressure Plates, the wooden housing for the Stepper, and the wooden brackets for the Pressure Plates. We also needed a few various electrical components, the solenoid valve, and the Waffler Iron itself.

The complete list of parts we used are listed below:

Major Parts

- Brand New Waffle Iron
- Solenoid Valve with 3/4" Metal Annulus + 3/4" pipe
- Large Plastic Bowl
- 80/20
- Wood
- Waffle Batter + PAM spray (and Water)

Smaller Parts

- HX711 Amplifier with LoadCell
- Songle Power Relay
- L298N Motor Shield
- One Arduino Uno + 4 mini-Breadboards
- M5 screws + M5 Nuts + M5 washers + Metal Brackets
- Stepper Motor
- 3D printed Plastic Pully + Single Braid Rope
- Wires

- Button

The only items we needed to buy for this project were the Waffle Iron, and the Solenoid Valve (thank your Professor Arnold). All of the 80/20 and its hardware components were given to us by David Lesser director of the MakerSpace at the Design and Innovation Building here on campus. David (Culver) brought the Plastic Bowl, the rope and the Waffle Batter alongside the PAM spray. The wood, metal annulus, and piper were provide courtesy of Professor Arnold. We both bought HX711 Amplifiers with 10kg Load Cells. All other component were provided by the PHYS 124 Lab.

We firmly believe our project has alot of reusabuility. All of the 80/20 and the connections can be re-purposed very easily for future projects. All of the hardware can be resused in a future project including the valve and pipe connections. The wood can always be resused, and the electrical components too. The only really unusable part again could be the things we used to make waffles, but this was used to feed the class.

XI. Expansion Options

An additional piece we could add on would to add in some bells and whistles. This could include an attached LCD screen that displays how many waffles have been made during this cooking cycle. We would need an additional Arduino, but it would be a fun thing to have add onto it.

Another fun addition we could add is a having a speaker play a fun song when the waffle finishes cooking to get you attention. This could make the machine even more fun to watch operate.

A very difficult project we could try and figure out is adding in a device that can pick the waffle out of the waffle iron and place on a plate next to the machine. This would be extremely difficult, as we would practically be making a robot arm to pick up waffles, however it would be really cool to see in action.

Why this fork would be so difficult is for many facets. First supporting the fork would be difficult. It would be hard to implement the forking mechanism with two degrees of freedom, that can retrieve a waffle, drop a waffle, and return to its original position, with good enough timing and can account for the variations of waffles made by our Waffler.

Further, the actual controls of the mechanism would be extremely difficult for us to use. We would have a lengthy job fo trying to write Arduino code to put the for in a specifies position each time. We would have to account for how much one step in one motor affects the other, and the path which the motor head takes. Since there are hard boundaries in the way, the motors would actually have to work synchronously to not collide with the wall of the machine. Further the inherent motor slop would make the repeatability a genuine concern of ours.

XII. De scoping Options

Instead of creating a custom batter dispenser we could have a human pour the batter in and have the aurduino tell the human when to stop pouring. The arduino could do this communication by flashing an LED green when the correct weight of batter has been sensed, and/or via a printing onto an LCD screen and communicating that enough batter has been poured. This is a dramatic descope, but the device would still be a fun thing to use as it could still function to open and close the waffle iron.

XIII. Lessons Learned

Probably the most important lesson we learned when building was to go with the more robust option. We initially though about using PVC, but that most definitely would have made the project way harder then possible.

We where pretty lucky because every competent we used worked as intended and we actually, so we didn't have any difficulties there.

XIV. Project Deliverables

At then end of the class we have created a video demonstrating the functionality of the Automatic Waffler. This video demonstrates how the waffler functions. Further this video illustrates all design features of the waffler, where we discuss about frame design, motor choice, sensing capabilities, and how we wired all the pieces to an Arduino.

In our Mechanical Considerations section we discuss many features of the waffler. The video will aim to go over each of these features is full and give an accurate account of there function.

We also gave an in person demonstration of the waffler highlighting it's key features and explaining how it works. This demo demonstrated the complete functionality of the Automatic Waffler, where we fed the entire class waffles.

XV. Appendix

- integrate: This is the Arduino code we used
- Total Construction: This is the final assembly we made in solid works.